# Osptek Display

# TFT LCD SPECIFICATION

Model No:

**YDP200H005-V9**

## REVISION RECORD

| REV NO. | REV DATE | CONTENTS | REMARKS |
|---|---|---|---|
| 0.1 | 2024-05-03 | First Release | Preliminary |
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## GENERAL INFORMATION

| Item | Contents | Unit |
|---|---|---|
| LCD Type | TFT | / |
| Viewing direction | ALL | O' Clock |
| Module outline (W × H) | 45.8 × 40.2×2.2 | mm |
| Active area (W×H) | 40.8 × 30.6 | mm |
| Number of Dots | 320(RGB) × 240 | / |
| Colors | 262K | / |
| Backlight Type | LED | / |
| Module Power consumption | / | mw |
| Interface Type | MCU 8 bit / 4-wire SPI | / |
| Input voltage | 2.8 | V |

## ABSOLUTE MAXIMUM RATINGS

| Parameter | Symbol | Min | Max | Unit |
|---|---|---|---|---|
| Supply voltage for logic | $V_{CC}$、$IOV_{CC}$ | -0.3 | 4.6 | V |
| Input voltage | $V_{IN}$ | -0.3 | $IOV_{CC}$+0.3 | V |
| Operating temperature | $T_{OP}$ | -20 | 65 | ℃ |
| Storage temperature | $T_{ST}$ | -30 | 70 | ℃ |
| Humidity | RH | - | 90%(Max60°C) | RH |

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## ELECTRICAL CHARACTERISTICS

DC CHARACTERISTICS

| Parameter | Symbol | Min | Typ | Max | Unit |
|---|---|---|---|---|---|
| Supply voltage for logic | $V_{CC}$-$V_{SS}$ | 3 | 3.3 | 3.6 | V |
| Input Current | Idd | - | 20 | 25 | mA |
| Input voltage ' H ' level | $V_{IH}$ | 0.8 $IOV_{CC}$ | - | $IOV_{CC}$ | V |
| Input voltage ' L ' level | $V_{IL}$ | VSS | 0 | 0.2 $IOV_{CC}$ | V |
| Output voltage ' H ' level | $V_{OH}$ | 0.8 $IOV_{CC}$ | - | $IOV_{CC}$ | V |
| Output voltage ' L ' level | $V_{OL}$ | VSS | 0 | 0.2 $IOV_{CC}$ | V |

## TIMING OF POWER SUPPLY

PLEASE REFER TO THE DRIVER IC SPECIFICATION.

## BACKLIGHT CHARACTERISTICS

| Item | Symbol | Min. | Typ. | Max. | Unit | Condition |
|---|---|---|---|---|---|---|
| Forward voltage | Vf | 3 | 3.2 | 3.4 | V | If=80 mA |
| Luminance | Lv | | | | $cd/m^2$ | |
| Number of LED | - | 4 | | | Piece | - |
| Connection mode | P | Parallel | | | - | - |

# 4.0 OPTICAL SPECIFICATION

## 4.1 Overview

The test of Optical specifications shall be measured in a dark room (ambient luminance ≤1lux and temperature = 25±2℃) with the equipment of Luminance meter system (Goniometer system and TOPCON BM-5) and test unit shall be located at an approximate distance 50cm from the LCD surface at a viewing angle of θ and Φ equal to 0°. The center of the measuring spot on the Display surface shall stay fixed.

The backlight should be operating for 30 minutes prior to measurement.

## 4.2 Optical Specifications

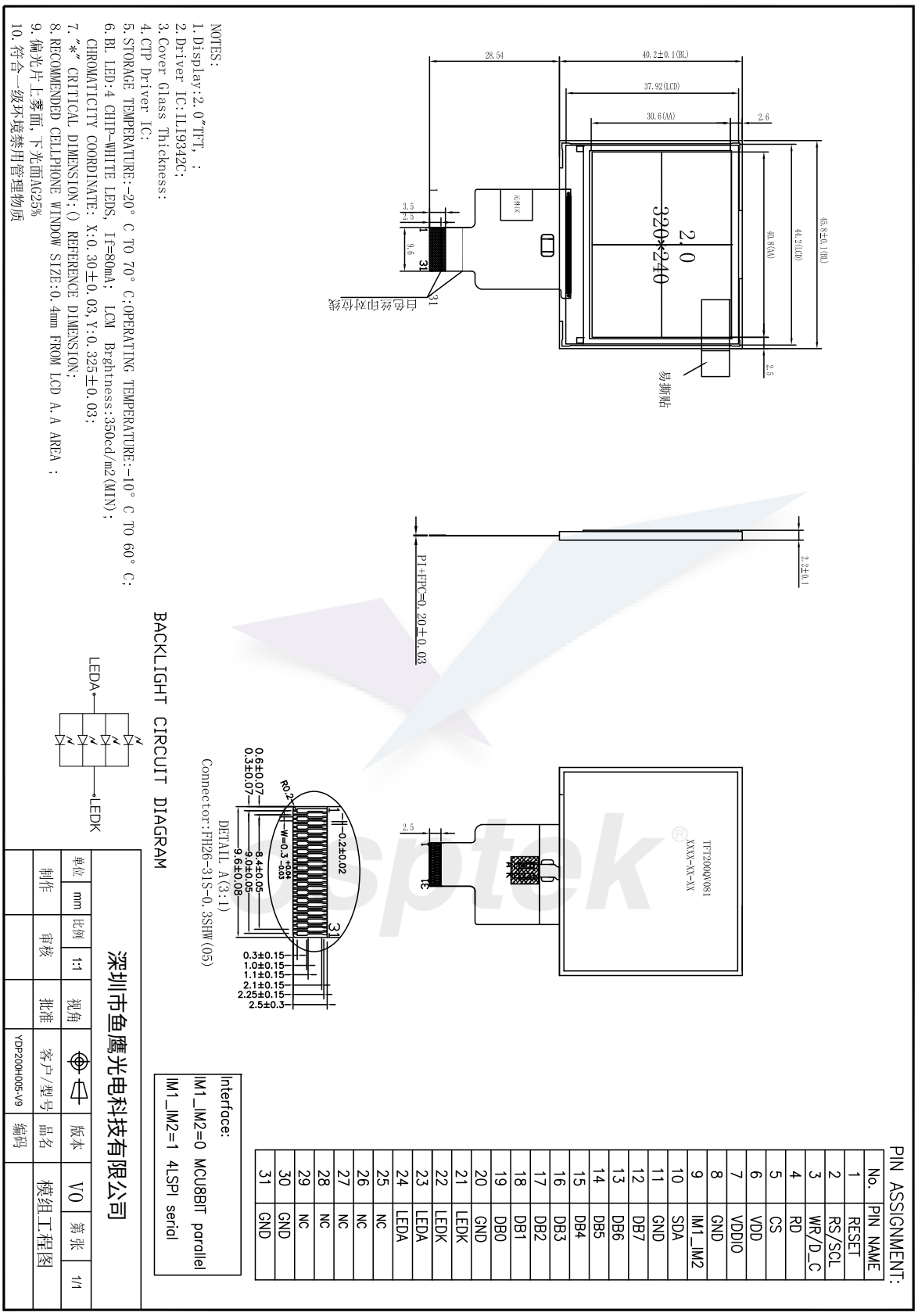
**<Table 4. Optical Specifications >**

| Parameter | | Symbol | Condition | Min. | Typ. | Max. | Unit | Remark |
|---|---|---|---|---|---|---|---|---|
| Viewing Angle Range | Horizontal | Θ3 | CR>10 | 70 | 80 | - | Deg. | Note1 |
| | | Θ9 | | 70 | 80 | - | Deg. | |
| | Vertical | Θ12 | | 70 | 80 | - | Deg. | |
| | | Θ6 | | 70 | 80 | - | Deg. | |
| Contrast ratio | | CR | Θ = 0° | 1000 | 1500 | - | | Note2 |
| Transmittance | | Tr | | 4.34 | 5.1 | | % | Note3 |
| Color Gamut | | CG | | 55 | 60 | | % | |
| Reproduction of color | Red | Rx | Θ = 0° | 0.618 | 0.638 | 0.658 | | Note4 (Based on C Light) |
| | | Ry | | 0.318 | 0.338 | 0.358 | | |
| | Green | Gx | | 0.276 | 0.296 | 0.326 | | |
| | | Gy | | 0.555 | 0.575 | 0.595 | | |
| | Blue | Bx | | 0.117 | 0.137 | 0.157 | | |
| | | By | | 0.104 | 0.124 | 0.144 | | |

| Parameter | Symbol | Condition | Min. | Typ. | Max. | Unit | Remark |
|---|---|---|---|---|---|---|---|
| White Chromaticity | Wx | Θ = 0° | 0.298 | 0.318 | 0.338 | | |
| | Wy | | 0.321 | 0.341 | 0.361 | | |
| Response Time (Rising + Falling) | $T_r + T_f$ | Ta= 25° C Θ = 0° | | 30 | 40 | ms | Note 5 |

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# EXTERNAL DIMENSIONS



## INTERFACE DESCRIPTION

| Pin.No | Symbol | Function |
|---|---|---|
| 1 | RESET | A reset pin |
| 2 | RS/SCL | -Display data/command selection pin in parallel interface<br>-This pin is used to be serial interface clock.(SCL) |
| 3 | WR/D-C | -Write enable in MCU parallel interface.<br>- Display data/command selection pin in 4-line serial interface. |
| 4 | RD | Read signal |
| 5 | CS | chip select signal input |
| 6 | VDD | Power supply(2.8v) |
| 7 | VDDIO | power supply（1.8V --2.8V） |
| 8 | GND | Ground |
| 9 | IM1_IM2 | interface mode select<br>IM1_IM2=0, 80-8bit parallel<br>IM1_IM2=1, 4-wire SPI |
| 10 | SDA | SPI interface input/output pin |
| 11 | GND | Ground |
| 12`19 | DB7`DB0 | Data bus |
| 20 | GND | Ground |
| 21 | LEDK | Back light power supply negative |
| 22 | LEDK | Back light power supply negative |
| 23 | LEDA | Back light power supply positive |
| 24 | LEDA | Back light power supply positive |
| 25 | NC | No connect |
| 26 | NC | No connect |
| 27 | NC | No connect |
| 28 | NC | No connect |
| 29 | NC | No connect |
| 30 | GND | Ground |
| 31 | GND | Ground |

## APPLICATION CIRCUIT

Please consult our technical department for detail information.

## INITIAL CODE

Please consult our technical department for detail information.

## RELIABILITY TEST

| No. | Test Item | Test Condition | Inspection after test |
|---|---|---|---|
| 1 | High Temperature Storage | 80±2℃/200 hours | Inspection after 2~4hours storage at room temperature, the sample shall be free from defects:<br>1.Air bubble in the LCD;<br>2.Sealleak;<br>3.Non-display;<br>4.missing segments;<br>5.Glass crack;<br>6.Current Idd is twice higher than initial value. |
| 2 | Low Temperature Storage | -30±2℃/200 hours | |
| 3 | High Temperature Operating | 70±2℃/120 hours | |
| 4 | Low Temperature Operating | -20±2℃/120 hours | |
| 5 | Temperature Cycle | -20±2℃~25~70±2℃×10cycles<br>(30min.) (5min.) (30min.) | |
| 6 | Damp Proof Test | 50℃±5℃×90%RH/120 hours | |
| 7 | Vibration Test | Frequency：10Hz~55Hz~10Hz<br>Amplitude：1.5mm,<br>X，Y，Z direction for total 3hours<br>(Packing condition) | |
| 8 | Dropping test | Drop to the ground from 1m height, one time, every side of carton.<br>(Packing condition) | |
| 9 | ESD test | Voltage:±8KV R: 330Ω C: 150pF<br>Air discharge, 10time | |

Remark:
1.The test samples should be applied to only one test item.
2.Sample size for each test item is 5~10pcs.
3.For Damp Proof Test, Pure water(Resistance＞10MΩ) should be used.
4.In case of malfunction defect caused by ESD damage, if it would be recovered to normal state after resetting, it would be judge as a good part.
Using ionizer(an antistatic blower) is recommended at working area in order to reduce electro-static voltage.
When removing protection film from LCM panel, peel off the tag slowly( recommended more than one second) while blowing with ionizer toward the peeling face to minimize ESD which may damage electrical circuit.
5.EL evaluation should be excepted from reliability test with humidity and temperature: Some defects such as black spot/blemish can happen by natural chemical reaction with humidity and Fluorescence EL has.
6.Please use automatic switch menu(or roll menu) testing mode when test operating mode

# INSPECTION CRITERION

This specification is made to be used as the standard acceptance/rejection criteria for Color mobile phone LCM.

1 Sample plan

Sampling plan according to GB/T2828.1-2003/ISO 2859-1： 1999 and ANSI/ASQC Z1.4-1993, normal level 2 and based on:

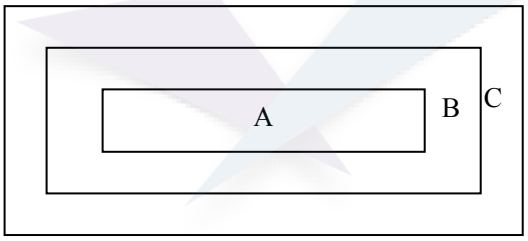
Major defect: AQL 0.65

Minor defect: AQL 1.5

2. Inspection condition

Viewing distance for cosmetic inspection is about 30cm with bare eyes, and under an environment of 20~40W light intensity, all directions for inspecting the sample should be within 45° against perpendicular line.

3. Definition of inspection zone in LCD.




Zone A: character/Digit area

Zone B: viewing area except Zone A (ZoneA+ZoneB=minimum Viewing area)

Zone C: Outside viewing area (invisible area after assembly in customer's product)

Fig.1 Inspection zones in an LCD.

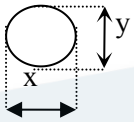
Note: As a general rule, visual defects in Zone C are permissible, when it is no trouble for quality and assembly of customer's product.

# INSPECTION STANDARDS

### 4.1 Major Defect

| Item No | Items to be inspected | Inspection Standard | Classification of defects |
|---|---|---|---|
| 4.1.1 | All functional defects | 1) No display<br>2) Display abnormally<br>3) Missing vertical， horizontal segment<br>4) Short circuit<br>5) Back-light no lighting, flickering and abnormal lighting. | Major |
| 4.1.2 | Missing | Missing component | |
| 4.1.3 | Outline dimension | Overall outline dimension beyond the drawing is not allowed. | |

### 4.2 Cosmetic Defect

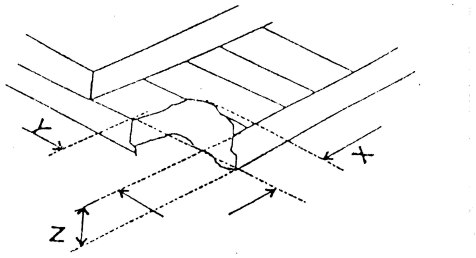
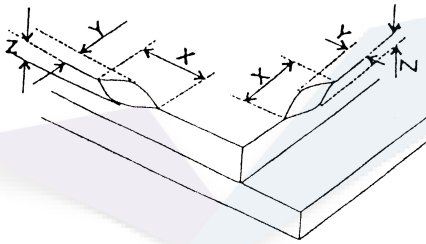
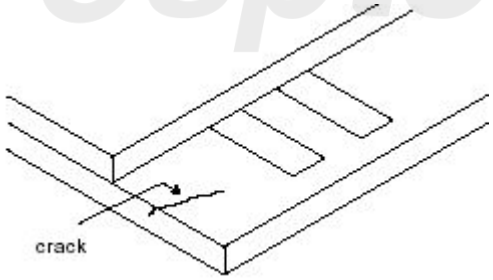
<table>
<tr><th>Item No</th><th>Items to be inspected</th><th>Inspection Standard</th><th>Classification of defects</th></tr>
<tr><td rowspan="2">4.2.1</td><td>Clear Spots<br><br>Black and white Spot defect Pinhole, Foreign Particle, Dirt under polarizer</td><td>For dark/white spot, size Φ is defined as $\Phi = \frac{(x+y)}{2}$<br>  <br><br>1.
<table>
<tr><th rowspan="2">Zone<br>Size(mm)</th><th colspan="3">Acceptable Qty</th></tr>
<tr><th>A</th><th>B</th><th>C</th></tr>
<tr><td>Φ ≤0.10</td><td colspan="2">Ignore</td><td rowspan="4">Ignore</td></tr>
<tr><td>0.10＜Φ≤0.15</td><td colspan="2">2</td></tr>
<tr><td>0.15＜Φ≤0.20</td><td colspan="2">1</td></tr>
<tr><td>Φ＞0.20</td><td colspan="2">0</td></tr>
</table></td><td>Minor</td></tr>
<tr><td>Dim Spots<br><br>Circle shaped and dim edged defects</td><td>2.
<table>
<tr><th rowspan="2">2. Zone<br>Size(mm)</th><th colspan="3">Acceptable Qty</th></tr>
<tr><th>A</th><th>B</th><th>C</th></tr>
<tr><td>Φ ≤0.2</td><td colspan="2">Ignore</td><td rowspan="4">Ignore</td></tr>
<tr><td>0.20＜Φ≤0.40</td><td colspan="2">3</td></tr>
<tr><td>0.40＜Φ≤0.60</td><td colspan="2">2</td></tr>
<tr><td>0.60＜Φ≤0.80</td><td colspan="2">1</td></tr>
<tr><td>0.80＜Φ</td><td colspan="2">0</td><td></td></tr>
</table></td><td>Minor</td></tr>
</table>

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4.2. Cosmetic Defect

<table>
<tr><th>Item No</th><th>Items to be inspected</th><th>Inspection Standard</th><th>Classification of defects</th></tr>
<tr><td>4.2.2</td><td>Line defect<br>Black line,<br>White line,<br>Foreign material under polarizer,</td><td>
<table>
<tr><th colspan="2">Size(mm)</th><th colspan="3">Acceptable Qty</th></tr>
<tr><th rowspan="2">L(Length)</th><th rowspan="2">W(Width)</th><th colspan="3">Zone</th></tr>
<tr><th>A</th><th>B</th><th>C</th></tr>
<tr><td>Ignore</td><td>W≤0.02</td><td colspan="2">Ignore</td><td rowspan="4">Ignore</td></tr>
<tr><td>L≤3.0</td><td>0.02＜W≤0.03</td><td colspan="2">2</td></tr>
<tr><td>L≤2.0</td><td>0.03＜W≤0.05</td><td colspan="2">1</td></tr>
<tr><td></td><td>0.05＜W</td><td colspan="2">Define as spot defect</td></tr>
</table>
</td><td>Minor</td></tr>
<tr><td>4.2.3</td><td>Polarizer scratch</td><td>If the Polarizer scratch can be seen after mobile phone cover assembling or in the operating condition, judge by the line defect of 4.2.2.<br>If the Polarizer scratch can be seen only in non-operating condition or some special angle, judge by the following.
<table>
<tr><th colspan="2">Size(mm)</th><th colspan="3">Acceptable Qty</th></tr>
<tr><th rowspan="2">L(Length)</th><th rowspan="2">W(Width)</th><th colspan="3">Zone</th></tr>
<tr><th>A</th><th>B</th><th>C</th></tr>
<tr><td>Ignore</td><td>W≤0.03</td><td colspan="2">Ignore</td><td rowspan="4">Ignore</td></tr>
<tr><td>5.0＜L≤10.0</td><td>0.03＜W≤0.05</td><td colspan="2">2</td></tr>
<tr><td>L≤5.0</td><td>0.05＜W≤0.08</td><td colspan="2">1</td></tr>
<tr><td></td><td>0.08＜W</td><td colspan="2">0</td></tr>
</table>
</td><td>Minor</td></tr>
<tr><td>4.2.4</td><td>Polarize<br>Air bubble</td><td>Air bubbles between glass & polarizer
<table>
<tr><th>2. Zone<br>Size(mm)</th><th colspan="3">Acceptable Qty</th></tr>
<tr><th></th><th>A</th><th>B</th><th>C</th></tr>
<tr><td>Φ≤0.2</td><td colspan="2">Ignore</td><td rowspan="4">Ignore</td></tr>
<tr><td>0.20＜Φ≤0.30</td><td colspan="2">2</td></tr>
<tr><td>0.30＜Φ≤0.50</td><td colspan="2">1</td></tr>
<tr><td>0.50＜Φ</td><td colspan="2">0</td></tr>
</table>
</td><td>Minor</td></tr>
</table>

4.3. Cosmetic Defect

<table>
<tr><th>Item No</th><th>Items to be inspected</th><th>Inspection Standard</th><th>Classification of defects</th></tr>
<tr><td rowspan="3">4.3.5</td><td rowspan="3">Glass defect</td><td>(i) Chips on corner<br>  <br><br><table><tr><th>X</th><th>Y</th><th>Z</th></tr><tr><td>≤2.0</td><td>≤S</td><td>Disregard</td></tr></table>Notes: S=contact pad length<br>Chips on the corner of terminal shall not be allowed to extend into the ITO pad or expose perimeter seal.</td><td>Minor</td></tr>
<tr><td>(ii)Usual surface cracks<br>  <br><br><table><tr><th>X</th><th>Y</th><th>Z</th></tr><tr><td>≤3.0</td><td><Inner border line of the seal</td><td>Disregard</td></tr></table></td><td>Minor</td></tr>
<tr><td>(iii) Crack<br>Cracks tend to break are not allowed.<br>  <br></td><td>Major</td></tr>
<tr><td>4.3.6</td><td>Parts alignment</td><td>1) Not allow IC and FPC/heat-seal lead width is more than 50% beyond lead pattern.<br>2) Not allow chip or solder component is off center more than 50% of the pad outline.</td><td>Minor</td></tr>
<tr><td>4.3.7</td><td>SMT</td><td>According to the <Acceptability of electronic assemblies> IPC-A-610C class 2 standard. Component missing or function defect are Major defect, the others are Minor defect.</td><td></td></tr>
</table>

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